

Submission STAT1-QUIZ-SUB01

Question STAT1-QUIZ-Q01

scratch: show step

1. $\bar{x} = 39/5 = 7.80$

2. $s^2 = \sum(x_i - \bar{x})^2 / (5 - 1) = 9.70$

=> FINAL: mean=7.80; sample variance=9.70 (check units/interpretation)

Question STAT1-QUIZ-Q02

scratch: show step

1. $P(D_1 \cap D_2) = (2/14)((2-1)/(14-1))$

2. =0.0110; the population shrinks after draw 1

=> FINAL: 0.0110 (check units/interpretation)

Question STAT1-QUIZ-Q03

scratch: show step

1. $SE = \sqrt{p(1-p)/n} = \sqrt{0.35(1-0.35)/40} = 0.0754$

2. $CI = 0.35 \pm 1.96(0.0754) = (0.202, 0.498)$

3. In repeated samples, the procedure captures the population proportion about 95% of the time.

=> FINAL: (0.202, 0.498); repeated-sampling interpretation (check units/interpretation)

Question STAT1-QUIZ-Q04

scratch: show step

1. Increase sample size n to reduce standard error.

2. Reduce confidence level or population variability when justified.

3. Lower confidence changes coverage, so precision is not free.

=> FINAL: increase n; lower confidence level or variability; note the coverage trade-off (check units/interpretation)

Question STAT1-QUIZ-Q05

scratch: show step

1. $0.03 < 0.05$, so reject H_0 under the stated rule.

2. Conclude evidence is inconsistent with H_0 ; do not say the probability H_0 is 3%.

=> FINAL: reject H_0 ; evidence against H_0 , not proof that H_0 is false (check units/interpretation)